

HUAXUAN (JASON) YANG

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EDUCATION

9/2022 - 6/2026 **University of California, Santa Cruz**

University

- Pursuing a major in Computer Engineering with Digital Hardware specialization
- 3.8 GPA
- Concentrated in hardware related courses: computer architecture, circuit analysis signals and systems, and PCB circuit design for high-speed logic. Also took research-oriented course in neural networks
- Took graduate course in RTL design and functional verification for ASICs/FPGAs
- Currently enrolled in a VLSI and chip design capstone course

EXPERIENCE

3/2025 - **Undergraduate Researcher**

Neuromorphic Computing Group

- Working with a team of neural network researchers and embedded engineers to develop a multimodal, real-time, object detection system under low lighting conditions via a fusion of RGB and Event camera frames
- Architected a custom, power and cost optimized hardware architecture for a real-time object detection system for an autonomous vehicle. Targeted high computational efficiency under a constrained power requirement via a specialized ASIC accelerator (Coral)
- Optimized neural network models (using TensorFlow/LiteRT) through compilation and quantization for efficient hardware acceleration on a Coral Edge TPU (Coral) to maximize accelerator utilization
- Met bi-weekly to cooperate with a system architect peer and model researchers to discuss architecture and implementation, drafting comprehensive hardware architecture documentation and contributing to a technical paper submission.

9/2024 - **Robotics Software Lead**

Slugbotics

- Led software subteam for an underwater Remotely Operated Vehicle (ROV), overseeing the design, development, and integration of the complete control stack.
- Developed topside control station GUI using Qt to provide operators with real-time telemetry monitoring and controls.
- Designed a low-latency camera stream solution for providing real-time visual feedback essential for remote piloting
- Designed and implemented the ROV's onboard computing system for handling sensor data acquisition, motor controls, and topside communication
- Collaborated closely with mechanical and electrical subteams to integrate software for hardware components, including motors, thrusters, and various sensors

6/2023 - 8/2024 **Software Developer Lead**

Pilotcity

- Led a frontend development intern team to develop and maintain an interactive internship platform using VueJS.
- Developed integrations with third-party APIs (including Videoask and Miro) to embed external content and activities within the main Vue application
- Mentored a team of junior developers on good software engineering practices, such as Git version control, Docker containerization, and performing code reviews
- Directly collaborated with artists and game designers to implement gamification features to increase user engagement
- Utilized various JavaScript web development frameworks, primarily VueJS

SKILLS

Computer Systems - Strong experience in interfacing with GNU/Linux and general UNIX systems. Proficient in various Linux kernel APIs and its various subsystems, such as computer networking in Linux (sockets, routing, packet filter, etc.), namespaces, etc.

Digital Hardware - Proficient in designing PCBs for high-speed logic with KiCAD. Proficient in debugging and testing with various test instruments (i.e., multimeters, oscilloscopes, etc.). Also adept at using open-source synthesis tools, RTL design, and functional verification with Verilog for FPGAs.

Machine Learning - Proficient in training and tuning models for computer vision.

Frontend Development - Proficient in designing web frontends using various JS frameworks

Programming - Proficient in C/C++, as well as Go. Proficient in various scripting languages such as Python, JavaScript, Shell, Tcl, etc. Familiar with Rust.